

# WorldSkills ASEAN 2025

## Network Systems Administration Training Journal

Version 1 - Day 1 to Day 3.2

|                         |                                                                |
|-------------------------|----------------------------------------------------------------|
| <b>Competitor</b>       | Theo Mata                                                      |
| <b>School/Context</b>   | MSU-IIT / WorldSkills Training                                 |
| <b>Track</b>            | IT Network Systems Administration                              |
| <b>Lab Platform</b>     | Proxmox VE Homelab                                             |
| <b>Document Purpose</b> | Personal review, GitHub portfolio, and competition preparation |

### Training Journal Note

This document summarizes the work completed from Day 1 through Day 3.2 of the WorldSkills Network Systems Administration preparation. It focuses on what was configured, why it mattered, what broke, how it was investigated, and what remains to be completed.

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# 1. Training Overview

This journal records the first phase of Theo Mata's preparation for the WorldSkills ASEAN 2025 IT Network Systems Administration task. The training started from basic Linux and networking concepts, then progressed into a working Proxmox-based homelab with virtualized servers, firewalling, DNS, DHCP, and DDNS troubleshooting.

## Main Training Goal

Build enough practical confidence to understand enterprise-level services instead of only copying commands. Every configuration is treated as a chance to learn packet flow, service roles, and troubleshooting methodology.

## Competition Context

The target module being practiced is Module A style Linux/network administration. The major services include an internal server named int-srv01.int.worldskills.ph, a firewall host fw.worldskills.ph, DNS, DHCP with DDNS, LDAP, CA, Samba, rsyslog, firewall rules, VPN, and DMZ services.

## Hardware Used

| Area             | Status  | Notes                                                                                      |
|------------------|---------|--------------------------------------------------------------------------------------------|
| Main workstation | Working | ASUS TUF A15 used for browser access, terminal work, documentation, and remote management. |
| Homelab server   | Working | Acer Aspire 5 running Proxmox VE headless.                                                 |
| Networking       | Working | Home LAN with Proxmox bridge vmbbr0 and virtual machines attached to the lab network.      |

## Core Learning Approach

- Read the requirement.
- Build the minimum working version.
- Verify with commands, not assumptions.
- Break problems into layers: link, IP, routing, DNS, service, firewall.
- Document the mistake and the fix.

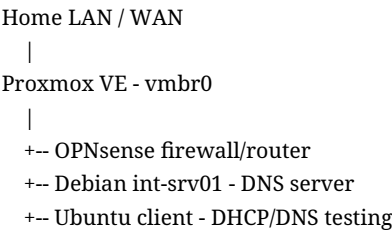
## 2. Homelab Architecture

The homelab uses Proxmox VE as the virtualization platform. The server is managed through the Proxmox web interface at 192.168.1.50:8006. The main virtual bridge used during the initial lab was vmbr0.

### Physical and Virtual Roles

| Area                 | Status                      | Notes                                                                                |
|----------------------|-----------------------------|--------------------------------------------------------------------------------------|
| Proxmox host         | Complete                    | Headless Acer Aspire 5. Physical NIC attached to vmbr0.                              |
| OPNsense VM          | Complete for training phase | Used as firewall/router and initially used for DHCP/Kea experiments.                 |
| Debian VM: int-srv01 | Working                     | Primary internal server for BIND9 DNS and future LDAP, CA, DHCP/DDNS, Samba, Syslog. |
| Ubuntu VM: client    | Working                     | Used as test client for DHCP, DNS resolution, and troubleshooting.                   |

Current high-level topology during the early lab:



### Important Addresses Used

| Area             | Status                     | Notes                                            |
|------------------|----------------------------|--------------------------------------------------|
| Proxmox host     | 192.168.1.50/24            | Management UI on https://192.168.1.50:8006       |
| OPNsense INT/LAN | 10.1.10.1/24               | Gateway used during initial INT network testing. |
| Debian int-srv01 | 10.1.10.10/24              | BIND9 DNS server.                                |
| Ubuntu client    | 10.1.10.100 or 10.1.10.175 | DHCP-assigned during testing.                    |

#### Architecture Correction Discovered Later

The initial working lab used OPNsense as DHCP server. After reviewing the actual module and marking sheet, DHCP with DDNS should be implemented under int-srv01, while the firewall should act as DHCP relay. This is now the next correction for Day 3.2.

### 3. Day 1 - Linux and Network Fundamentals

Day 1 focused on understanding the basic commands and concepts needed before configuring enterprise services. The main goal was not to memorize commands, but to understand what each command reveals about the system.

#### Concepts Covered

- IPv4 addressing and subnet masks
- Default gateway and routing table
- Linux network interfaces
- DNS resolution basics
- ICMP testing with ping
- Service status using systemctl
- Listening ports using ss
- Basic troubleshooting order

#### Important Commands Practiced

```
ip a
ip route
ping 8.8.8.8
ping google.com
ss -tulnp
systemctl status <service>
journalctl -u <service>
cat /etc/resolv.conf
resolvectl status
```

#### Key Lessons

- An IP address alone is not enough; routing and DNS must also work.
- If ping to IP works but ping to domain fails, suspect DNS.
- If a service is installed but not listening on a port, check systemctl and logs.
- Troubleshooting should follow evidence instead of guessing.

#### Day 1 Outcome

| Area                     | Status   | Notes                                                          |
|--------------------------|----------|----------------------------------------------------------------|
| Linux command confidence | Improved | Able to inspect IP, routes, DNS, listening services, and logs. |
| Networking fundamentals  | Improved | Understood the role of gateway, DNS server, and route table.   |
| Troubleshooting mindset  | Started  | Began using layered investigation instead of random fixes.     |

## 4. Day 2 - Infrastructure Deployment

Day 2 focused on building the virtual infrastructure needed for the WorldSkills-style lab. This included creating virtual machines, understanding Proxmox networking, and preparing Debian, Ubuntu, and OPNsense systems.

### Proxmox VE

Proxmox VE became the core hypervisor for the training environment. The Proxmox host uses vmbr0 as the main Linux bridge connected to the physical network.

Proxmox access:

`https://192.168.1.50:8006`

Main bridge:

`vmbr0`

Physical interface:

`nic0` / physical Ethernet adapter

### VMs Created

| Area          | Status                  | Notes                                                                  |
|---------------|-------------------------|------------------------------------------------------------------------|
| Ubuntu Server | Created                 | Used as DHCP/DNS client testing machine.                               |
| Debian Server | Created                 | Became int-srv01 for BIND9 and future internal services.               |
| OPNsense      | Created and reinstalled | Used for firewall/router practice and early Kea DHCP/DDNS experiments. |

### OPNsense Installation and Interface Mapping

OPNsense was installed in Proxmox using UEFI/OVMF and VirtIO network adapters. The firewall VM was assigned multiple virtual NICs. Inside OPNsense, these appeared as vtnet interfaces.

Interface mapping practiced:

WAN -> vtnet0

LAN -> vtnet1

OPT1 -> vtnet2

OPT2 -> vtnet3

Example final early addressing:

LAN/INT: 10.1.10.1/24

OPT1: 10.1.20.1/24

OPT2: 10.1.40.1/24

WAN: DHCP from home network

### Connectivity Verification

From OPNsense console:

`ping 8.8.8.8`

`ping google.com`

Result:

0% packet loss during successful tests.

### Important Learning

Virtual interface names inside a VM do not directly show the Proxmox bridge name. The order of virtual NICs in Proxmox usually determines whether OPNsense sees them as vtnet0, vtnet1, and so on.

## 5. Day 3.1 - Enterprise DNS Foundation

Day 3.1 focused on installing and configuring BIND9 on Debian. This became the foundation for internal DNS, reverse DNS, and later DDNS.

### BIND9 Installation

```
apt update
apt install bind9 bind9-utils dnsutils -y
systemctl status named
ss -tulnp | grep named
```

### Initial Zone: worldskills.ph

The first working DNS zone used worldskills.ph. This was useful for learning BIND9 syntax, SOA records, NS records, A records, and PTR records.

```
zone "worldskills.ph" {
    type master;
    file "/etc/bind/db.worldskills.ph";
};

zone "10.1.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10.1.10";
};
```

### Forward Zone File

```
$TTL 604800
@ IN SOA ns1.worldskills.ph. admin.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@      IN NS ns1.worldskills.ph.
ns1    IN A  10.1.10.10
int-srv01 IN A 10.1.10.10
@      IN A  10.1.10.10
```

### Reverse Zone File

```
$TTL 604800
@ IN SOA ns1.worldskills.ph. admin.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@ IN NS ns1.worldskills.ph.
10 IN PTR int-srv01.worldskills.ph.
```

### Validation Commands

```
named-checkconf
named-checkzone worldskills.ph /etc/bind/db.worldskills.ph
```

```
named-checkzone 10.1.10.in-addr.arpa /etc/bind/db.10.1.10
systemctl reload named
```

```
dig @10.1.10.10 ns1.worldskills.ph
dig @10.1.10.10 int-srv01.worldskills.ph
dig @10.1.10.10 -x 10.1.10.10
```

## DNS Lessons Learned

- Forward DNS maps hostname to IP.
- Reverse DNS maps IP to hostname through PTR records.
- A missing trailing dot can accidentally duplicate domain suffixes.
- named-checkzone can say OK even when the record design is not ideal.
- dig, host, and nslookup each show DNS evidence differently.



## 6. Day 3.2 - WorldSkills DNS, DHCP and DDNS Alignment

Day 3.2 moved from a simple working DNS lab toward the actual WorldSkills requirement. The official internal zone is int.worldskills.ph, not just worldskills.ph. This day also introduced LDAP SRV records and DDNS troubleshooting.

### WorldSkills Requirement Interpreted

The module states that int-srv01.int.worldskills.ph acts as the internal server providing authentication, certificate authority, file and DNS services. The DHCP section is also under this internal server section, meaning DHCP with DDNS should be implemented on int-srv01, while the firewall handles relay/routing.

| Area           | Status   | Notes                                                                  |
|----------------|----------|------------------------------------------------------------------------|
| Internal DNS   | Required | Create int.worldskills.ph and reverse zones.                           |
| DNS records    | Required | Add records according to DNS zone notes.                               |
| Recursion      | Required | Employee devices use this DNS server.                                  |
| DHCP with DDNS | Required | Assign IPv4/IPv6 to int-client and register IPv4 via DDNS.             |
| Firewall role  | Separate | Firewall performs routing, NAT, filtering, VPN, proxy, and DHCP relay. |

### Created Official Zone: int.worldskills.ph

```
include "/etc/bind/ddns.key";

zone "int.worldskills.ph" {
    type master;
    file "/etc/bind/db.int.worldskills.ph";
    allow-update { key "ddns-key"; };
};

zone "10.1.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10.1.10";
};
```

### Current db.int.worldskills.ph Content

```
$TTL 604800
@ IN SOA ns1.int.worldskills.ph. admin.int.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )

@      IN NS ns1.int.worldskills.ph.
ns1    IN A  10.1.10.10
int-srv01 IN A 10.1.10.10
@      IN A  10.1.10.10
_auth._tcp IN SRV 10 50 389 int-srv01.int.worldskills.ph.
```

### Marking Sheet Correction

The marking sheet expects the SRV record \_ldap.\_tcp.auth.int.worldskills.ph. The current \_auth.\_tcp record worked technically, but the exact record name must be corrected for scoring.

## SRV Record Troubleshooting

The LDAP SRV record initially returned a duplicated target name because the SRV target lacked a trailing dot.

Wrong target result:

int-srv01.int.worldskills.ph.int.worldskills.ph.

Fixed record:

\_auth.\_tcp IN SRV 10 50 389 int-srv01.int.worldskills.ph.

Verification:

dig @10.1.10.10 SRV \_auth.\_tcp.int.worldskills.ph

## Early DHCP Integration Using OPNsense

OPNsense Kea DHCP was initially configured to provide IPv4 leases on the 10.1.10.0/24 lab network. The DHCP options delivered DNS server 10.1.10.10 and search domain int.worldskills.ph to the Ubuntu client.

Example early DHCP settings:

Subnet: 10.1.10.0/24

Pool: 10.1.10.100 - 10.1.10.200

Gateway: 10.1.10.1

DNS server: 10.1.10.10

Search domain: int.worldskills.ph

## Dnsmasq Conflict Found and Fixed

A major troubleshooting step was discovering that Dnsmasq DNS & DHCP was still enabled in OPNsense. This caused the client to receive the wrong DNS server.

Problem observed on Ubuntu:

resolvectl dns -> 10.1.10.1

resolvectl domain -> internal

Expected:

DNS server -> 10.1.10.10

Search domain -> worldskills.ph / int.worldskills.ph

Root cause:

Dnsmasq and Kea DHCP were conflicting.

Fix:

Disable Dnsmasq DNS & DHCP.

Use Kea DHCP only during the early lab.

## DDNS Attempt Through OPNsense

A TSIG key was created and BIND was configured to accept dynamic updates. OPNsense Kea DHCP/DDNS Agent was also configured. Later review showed the module expects DHCP/DDNS to run on int-srv01, so this OPNsense path is now considered a learning exercise rather than the final scoring architecture.

DDNS components tested:

- /etc/bind/ddns.key

- allow-update { key "ddns-key"; };

- OPNsense Kea DHCPv4 Dynamic DNS fields

- OPNsense DDNS Agent on 127.0.0.1:53001

- BIND update test using nsupdate

## DDNS Troubleshooting Findings

| Area                          | Status               | Notes                                                                            |
|-------------------------------|----------------------|----------------------------------------------------------------------------------|
| OPNsense to BIND connectivity | Working              | nc and drill from OPNsense to 10.1.10.10:53 succeeded.                           |
| DDNS Agent                    | Working              | kea-dhcp-ddns started and listened on 127.0.0.1:53001 after fixing bind address. |
| TSIG key                      | Working              | nsupdate debug showed TSIG NOERROR.                                              |
| Manual nsupdate               | Failed with SERVFAIL | BIND received the update but could not apply it.                                 |
| Potential cause               | Investigating        | BIND update processing, write permissions, journal creation, or AppArmor.        |
| Architecture correction       | Important            | Final implementation should move DHCP/DDNS to Debian int-srv01.                  |

## 7. Troubleshooting Playbook

This section summarizes the troubleshooting habits developed during the first three days.

### DNS Troubleshooting Flow

1. Is the DNS service running?  
`systemctl status named`
2. Is it listening on port 53?  
`ss -tulnp | grep named`
3. Is the zone syntax valid?  
`named-checkconf`  
`named-checkzone <zone> <file>`
4. Does direct query to DNS server work?  
`dig @10.1.10.10 <name>`
5. Does the client use the correct DNS server?  
`resolvectl dns`  
`cat /etc/resolv.conf`
6. Does the search suffix work?  
`resolvectl domain`  
`ping int-srv01`

### DHCP Troubleshooting Flow

1. Release and renew lease:  
`sudo dhclient -r ens18`  
`sudo dhclient -v ens18`
2. Confirm IP and gateway:  
`ip a`  
`ip route`
3. Inspect received DHCP options:  
`grep -E "domain-name|domain-name-servers" /var/lib/dhcp/dhclient*.leases`
4. Check for multiple DHCP services:  
Dnsmasq, Kea, ISC DHCP, router DHCP
5. Confirm the module-required server owns DHCP.

### DDNS Troubleshooting Flow

1. Confirm BIND accepts dynamic updates.
2. Confirm TSIG key is loaded by BIND.
3. Test manual update:  
`nsupdate -d -k /etc/bind/ddns.key`
4. Confirm .jnl journal can be created.
5. Check logs for named update errors.
6. Confirm DHCP server is the intended module server.
7. Confirm client hostname is being sent.
8. Verify forward and reverse registration.

## Biggest Troubleshooting Lesson

The most important discovery was not a command; it was architecture. A working setup is not always a scoring setup. The task places DHCP with DDNS under int-srv01, so the final implementation must match the module design.

## 8. Marking Sheet Alignment

The uploaded marking sheet clarified which items are likely scored. This section maps current progress against A2: int-srv01 DNS, DHCP and DDNS.

| Area                                | Status           | Notes                                                                       |
|-------------------------------------|------------------|-----------------------------------------------------------------------------|
| DNS A record for int-srv01          | Done             | int-srv01.int.worldskills.ph resolves to 10.1.10.10.                        |
| DNS PTR for int-srv01               | Done             | 10.1.10.10 reverse lookup resolves to int-srv01.int.worldskills.ph.         |
| DNS AAAA for int-srv01              | Pending          | Need to add 2001:db8:1001:10::10.                                           |
| SRV record exact name               | Partial          | Working SRV exists, but marking expects _ldap._tcp.auth.int.worldskills.ph. |
| Recursive queries                   | Done             | External domains resolved successfully during testing.                      |
| Secondary DNS for DMZ zone          | Pending          | Not started.                                                                |
| Secondary DNS for DMZ reverse zones | Pending          | Not started.                                                                |
| DHCP IPv4 and IPv6 for int-client   | Pending redesign | Initial OPNsense IPv4 scope worked, but module expects DHCP on int-srv01.   |
| DDNS IPv4 dynamic updates           | In progress      | TSIG/BIND/DDNS explored; final version should be implemented on int-srv01.  |
| IPv6 connectivity to INT and DMZ    | Pending          | Requires network/VLAN/DMZ continuation.                                     |

### Quick Marks to Fix Next

10. Add AAAA record for int-srv01.
11. Add exact SRV record: \_ldap.\_tcp.auth.int.worldskills.ph.
12. Move DHCP/DDNS implementation to int-srv01.
13. Configure firewall as DHCP relay to int-srv01.
14. Configure DHCPv6 range for int-client.
15. Configure secondary DNS zones for dmz.worldskills.ph and reverse DMZ zones.

## 9. Current Status and Next Steps

### What Is Solid Already

- Proxmox VM infrastructure is functional.
- OPNsense installation and interface assignment were practiced.
- Debian BIND9 is installed and serving zones.
- Forward and reverse DNS concepts are understood.
- The official int.worldskills.ph zone exists.
- DNS validation commands are familiar.
- DHCP option troubleshooting was practiced deeply.
- SRV record troubleshooting was completed.
- DDNS architecture and TSIG concepts were introduced.

### What Needs Correction

- Do not rely on OPNsense as the final DHCP server for the module.
- Move DHCP with DDNS to int-srv01.
- Use firewall DHCP relay toward int-srv01.
- Correct SRV record naming according to the marking sheet.
- Add IPv6 records and DHCPv6 configuration.
- Continue with LDAP, CA, Samba, and Syslog after DNS/DHCP is aligned.

### Recommended Next Session Plan

Day 3.2 continuation plan:

1. Backup current BIND configuration.
2. Add missing AAAA record.
3. Add exact LDAP SRV record required by marking sheet.
4. Install Kea DHCP on int-srv01.
5. Configure IPv4 scope 10.1.40.100-199/24.
6. Configure IPv6 scope 2001:db8:1001:40::100-199/64.
7. Configure DDNS locally from Kea DHCP to BIND9.
8. Configure firewall DHCP relay to int-srv01.
9. Verify int-client receives IPv4 and IPv6.
10. Verify int-client registers in DNS via DDNS.

### Personal Progress Reflection

At the beginning of the training, the main challenge was understanding what enterprise DNS and DHCP even meant. By Day 3.2, the work had progressed into real troubleshooting: conflicting DHCP services, wrong DNS options, BIND zone syntax, missing trailing dots, TSIG keys, Kea DDNS agents, and interpreting the marking sheet. The biggest improvement is the ability to reason through the system instead of simply copying commands.

#### Version 1 Completion Note

This Training Journal Version 1 covers Day 1 through Day 3.2. It should be updated after DHCP/DDNS is rebuilt on int-srv01 and after the next module services are implemented.